

AI-Powered Outbound Calls (Arabic)

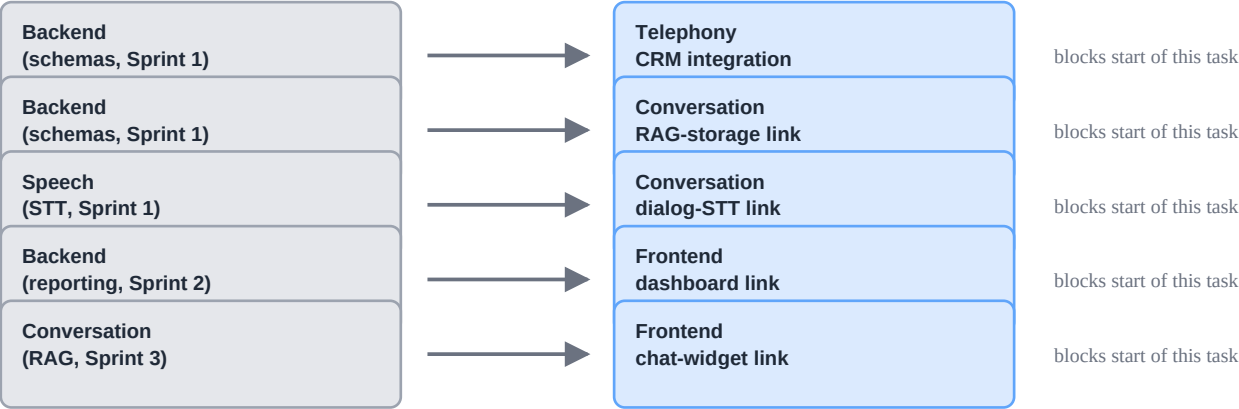
4-Sprint Plan — 1 Week per Sprint

Team size	5 members, one per module
Duration	4 weeks (4 sprints x 1 week)
Sequencing rule	Foundational/independent work first; dependent work scheduled after its blocker's sprint completes

Team & Color Legend

Person A	Person B	Person C	Person D	Person E
Speech Processing (STT/TTS)	Telephony & Call Orchestration	Conversation/NLU & RAG	Backend/Data & Reporting	Frontend/Dashboard

Cross-Team Dependency Chain



Sprint-by-Sprint Assignment Grid

Person A Speech Processing (STT/TTS)	- Set up Whisper API / self-hosted model - Test accuracy: Egyptian Arabic vs MSA <i>Working, validated Arabic STT — unblocks Conversation's dialog integration</i>	- Select TTS provider + Arabic voice samples - Generate dynamic greeting script playback <i>Natural Arabic TTS with dynamic call greetings</i>	Set up pydub/ffmpeg for call audio format handling <i>Reliable audio conversion between telephony and STT/TTS</i>	- Support full call-flow integration testing - Tune STT/TTS from real test-call results <i>Polish, bug fixes, accuracy tuning</i>
Person B Telephony & Call Orchestration	- Create Twilio/Vonage account, API keys - Test outbound dial <i>Confirmed ability to place outbound calls programmatically</i>	- Build retry logic (no-answer/failed call) - Build fallback-to-human-agent transfer <i>Resilient call handling — retries + live-agent escalation</i>	Build queue system with Celery + Redis <i>Scheduled/queued outbound calling working end-to-end</i>	- Integrate CRM/customer-flagged-list API (needs Backend, Sprint 1) - Support full call-flow integration testing <i>Scheduler pulls real follow-up lists from CRM</i>
Person C Conversation/NLU & RAG	Build branching dialog tree (yes/no/uncertain) <i>Core dialog logic working on placeholder STT input</i>	- Set up vector DB (Pinecone) - Build document ingestion/chunking/embedding job <i>KB documents embedded and searchable</i>	- Integrate GPT-4 Turbo answer generation with citations - Integrate dialog logic with finalized STT (needs Speech, Sprint 1) <i>RAG chatbot gives cited answers; dialog uses real STT output</i>	- Connect RAG pipeline to KB document storage (needs Backend, Sprint 1) - Support integration testing <i>Full RAG pipeline live against real KB storage</i>
Person D Backend/Data & Reporting	- Design call logs schema (PostgreSQL/MongoDB) - Design KB storage schema <i>Schemas ready — unblocks Telephony's CRM link + Conversation's RAG storage</i>	- Build First Call Resolutions report generator - Set up logging (ELK or Grafana+Loki) <i>Reporting pipeline live — unblocks Frontend's dashboard integration</i>	Implement OAuth2 / role-based access for KB data <i>Secure, auditable access to chatbot and internal data</i>	- Support integration testing - Fix schema/reporting issues found in testing <i>System hardening, bug fixes from integration feedback</i>

Person E Frontend/Dashboard	• Build React dashboard shell Base dashboard app structure ready	• Add charts: FCR rate / AHT / completion % • Build embeddable chat UI component Dashboard visuals + chat widget UI built (mock data)	• Integrate dashboard with reporting API (needs Backend, Sprint 2) Dashboard shows live data instead of mocks	• Connect chat widget to RAG API (needs Conversation, Sprint 3) • Write integration tests across full call flow • Write test suite for chatbot accuracy Full system connected end-to-end; test coverage in place

Visual Timeline — Who's Doing What, Each Week

	Sprint 1	Sprint 2	Sprint 3	Sprint 4
Person A Speech Processing (STT/TTS)	Set up Whisper API / self-hosted ... + 1 more task(s)	Select TTS provider + Arabic voic... + 1 more task(s)	Set up pydub/ffmpeg for call audi...	Support full call-flow integratio... + 1 more task(s)
Person B Telephony & Call Orchestra	Create Twilio/Vonage account, API... + 1 more task(s)	Build retry logic (no-answer/fail... + 1 more task(s)	Build queue system with Celery + ...	Integrate CRM/customer-flagged-li... + 1 more task(s) ■ has dependency
Person C Conversation/NLU & RAG	Build branching dialog tree (yes/...	Set up vector DB (Pinecone) + 1 more task(s)	Integrate GPT-4 Turbo answer gene... + 1 more task(s) ■ has dependency	Connect RAG pipeline to KB docume... + 1 more task(s) ■ has dependency
Person D Backend/Data & Reporting	Design call logs schema (PostgreS... + 1 more task(s)	Build First Call Resolutions repo... + 1 more task(s)	Implement OAuth2 / role-based acc...	Support integration testing + 1 more task(s)
Person E Frontend/Dashboard	Build React dashboard shell	Add charts: FCR rate / AHT / comp... + 1 more task(s)	Integrate dashboard with reportin... ■ has dependency	Connect chat widget to RAG API (n... + 2 more task(s) ■ has dependency

Light-colored blocks show each person's primary sprint focus; 'has dependency' flags a task that depends on another module's work completing in a prior sprint (see dependency chain, page 1).

Person A

Speech Processing (STT/TTS)

Sprint 1

- Set up Whisper API / self-hosted model
- Test accuracy: Egyptian Arabic vs MSA

Goal: *Working, validated Arabic STT — unblocks Conversation's dialog integration*

Sprint 2

- Select TTS provider + Arabic voice samples
- Generate dynamic greeting script playback

Goal: *Natural Arabic TTS with dynamic call greetings*

Sprint 3

- Set up pydub/ffmpeg for call audio format handling

Goal: *Reliable audio conversion between telephony and STT/TTS*

Sprint 4

- Support full call-flow integration testing
- Tune STT/TTS from real test-call results

Goal: *Polish, bug fixes, accuracy tuning*

Person B

Telephony & Call Orchestration

Sprint 1

- Create Twilio/Vonage account, API keys
- Test outbound dial

Goal: *Confirmed ability to place outbound calls programmatically*

Sprint 2

- Build retry logic (no-answer/failed call)
- Build fallback-to-human-agent transfer

Goal: *Resilient call handling — retries + live-agent escalation*

Sprint 3

- Build queue system with Celery + Redis

Goal: *Scheduled/queued outbound calling working end-to-end*

Sprint 4

- Integrate CRM/customer-flagged-list API (needs Backend, Sprint 1)
- Support full call-flow integration testing

Goal: *Scheduler pulls real follow-up lists from CRM*

Person C

Conversation/NLU & RAG

Sprint 1

- Build branching dialog tree (yes/no/uncertain)

Goal: *Core dialog logic working on placeholder STT input*

Sprint 2

- Set up vector DB (Pinecone)
- Build document ingestion/chunking/embedding job

Goal: *KB documents embedded and searchable*

Sprint 3

- Integrate GPT-4 Turbo answer generation with citations
- Integrate dialog logic with finalized STT (needs Speech, Sprint 1)

Goal: *RAG chatbot gives cited answers; dialog uses real STT output*

Sprint 4

- Connect RAG pipeline to KB document storage (needs Backend, Sprint 1)
- Support integration testing

Goal: *Full RAG pipeline live against real KB storage*

Person D

Backend/Data & Reporting

Sprint 1

- Design call logs schema (PostgreSQL/MongoDB)
- Design KB storage schema

Goal: *Schemas ready — unblocks Telephony's CRM link + Conversation's RAG storage*

Sprint 2

- Build First Call Resolutions report generator
- Set up logging (ELK or Grafana+Loki)

Goal: *Reporting pipeline live — unblocks Frontend's dashboard integration*

Sprint 3

- Implement OAuth2 / role-based access for KB data

Goal: *Secure, auditable access to chatbot and internal data*

Sprint 4

- Support integration testing
- Fix schema/reporting issues found in testing

Goal: *System hardening, bug fixes from integration feedback*

Person E

Frontend/Dashboard

Sprint 1

- Build React dashboard shell

Goal: *Base dashboard app structure ready*

Sprint 2

- Add charts: FCR rate / AHT / completion %
- Build embeddable chat UI component

Goal: *Dashboard visuals + chat widget UI built (mock data)*

Sprint 3

- Integrate dashboard with reporting API (needs Backend, Sprint 2)

Goal: *Dashboard shows live data instead of mocks*

Sprint 4

- Connect chat widget to RAG API (needs Conversation, Sprint 3)
- Write integration tests across full call flow
- Write test suite for chatbot accuracy

Goal: *Full system connected end-to-end; test coverage in place*